

Sprint 1 Bootstrap: 36-State Factored MDP with LLM-Estimated Transitions

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1 MDP Formulation

The Sprint 1 simulator implements a finite-horizon factored MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$ with 36 states, 4 actions, and transitions estimated by an LLM (deepseek-v4-flash) using Algorithm 2.

1.1 State Space

Four factors: step bin (`step_bin` $\in \{\text{inactive}, \text{moderate}, \text{active}\}$), sleep quality (`sleep` $\in \{\text{good}, \text{poor}\}$), day type (`day_of_week` $\in \{\text{weekday}, \text{weekend}\}$), and burden level (`burden` $\in \{\text{low}, \text{medium}, \text{high}\}$):

$$\mathcal{S} = \{\text{inactive}, \text{moderate}, \text{active}\} \times \{\text{good}, \text{poor}\} \times \{\text{weekday}, \text{weekend}\} \times \{\text{low}, \text{medium}, \text{high}\}$$

Three factors are deterministic: `day_of_week` cycles daily through a 7-day pattern; `burden` uses a rolling window of the last 3 actions (increments on non-idle actions). Two factors are stochastic: `step_bin` at each timestep, and `sleep` at day boundaries.

1.2 Action Space

$$\mathcal{A} = \{\text{idle}, \text{movement_suggestion}, \text{goal_reminder}, \text{journal}\}$$

All non-idle actions incur a penalty of 0.05.

1.3 Transition Function

Transitions are loaded from 6 JSON probability tables generated by an LLM (deepseek-v4-flash, $N = 16$ samples per state-action pair, Algorithm 2):

- `day_boundary.json`: $P(\text{sleep}' \mid \text{step_bin}, \text{burden}, \text{day_of_week}, \text{sleep})$ – action-independent, evaluated at step 0 of each day.
- `within_day_0.json` through `within_day_4.json`: $P(\text{step_bin}' \mid \text{step_bin}, \text{burden}, \text{action}, \text{day_of_week}, \text{sleep})$ – one table per timestep, action-conditioned.

The LLM’s estimates reflect plausible real-world dynamics: interventions have modest effects on step counts, and sleep quality influences next-day activity.

1.4 Reward

$$R = \alpha \cdot \text{step_bin_value} + (1 - \alpha) \cdot \text{sleep_value} - \text{action_penalty}$$

with $\alpha = 0.9$. Step bin values: `inactive` $\rightarrow$ 0.0, `moderate` $\rightarrow$ 0.5, `active` $\rightarrow$ 1.0. Sleep values: `good` $\rightarrow$ 1.0, `poor` $\rightarrow$ -1.0. Action penalty: 0.0 for idle, 0.05 otherwise.

Optional `reward_multiplier_by_step` masks the last step of each day: $[1, 1, 1, 1, 0]$.

1.5 Episode

$T = 90 \text{ days} \times 5 \text{ steps/day} = 450 \text{ timesteps per episode}$.

2 Reference Policy Bounds

Six reference policies evaluated via simulation (10 seeds, 450 timesteps):

Policy	Total Reward	Per Step	Description
Fixed idle	380.4 ± 3.7	0.845	Never intervene (no penalty)
Greedy oracle (1-step)	281.0 ± 20.9	0.625	Myopic expected-reward maximisation
Random	100.1 ± 11.7	0.222	Uniform action selection
Fixed movement_suggestion	35.2 ± 4.8	0.078	Always suggest movement
Fixed journal	50.5 ± 5.1	0.112	Always journal
Fixed goal_reminder	91.2 ± 7.1	0.203	Always remind of goal

Table 1: Reference policy bounds (10 seeds). Always-idle dominates because the LLM estimates natural activity is already high and action penalties are costly. The greedy oracle’s myopic one-step lookahead fails to capture long-term value.

Always-idle at 0.845/step sets an upper bound for agents: the LLM estimates that most users maintain moderate-to-active step counts without intervention, and paying a 0.05 penalty for no improvement is worse than doing nothing.

3 Base Benchmark Results

50 seeds, 450 timesteps, default hyperparameters (EG $\epsilon = 0.1$, UCB $c = 2.0$, TS $\alpha_0 = \beta_0 = 1$). Config: `configs/sprint1_bootstrap.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Epsilon-Greedy ($\epsilon = 0.1$)	280.8 ± 89.3	0.624	0.743
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	252.6 ± 105.6	0.561	0.662
UCB ($c = 2.0$)	240.4 ± 33.9	0.534	0.803
Random	99.7 ± 10.9	0.222	0.215

Table 2: Base benchmark results (50 seeds, 450 timesteps). Learning agents massively outperform random (2.5–2.8 $\times$), confirming the bootstrap tables contain exploitable structure. EG leads.

Key observation: under bootstrap transitions, learning agents achieve 2.5–2.8 $\times$ random, starkly different from the random-transition baseline where all agents clustered within 3% of each other. The LLM-estimated dynamics have meaningful structure.

4 Full Benchmark Results

50 seeds, 450 timesteps, 9 agents with MVP-optimised hyperparameters (EG $\epsilon = 0.05$, D-EG $\epsilon_{\text{start}} = 0.2$, $T_{\text{decay}} = 200$, UCB $c = 0.5$). Config: `configs/sprint1_bootstrap_extensions.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	358.1 ± 14.0	0.796	0.837
Standard D-EG	287.8 ± 106.9	0.640	0.718
Standard EG ($\epsilon = 0.05$)	282.3 ± 95.0	0.627	0.748
Contextual UCB	268.3 ± 29.6	0.596	0.733
Standard TS	252.6 ± 105.6	0.561	0.662
Contextual D-EG	240.4 ± 93.0	0.534	0.614
Contextual EG	216.9 ± 95.7	0.482	0.583
Contextual TS	155.5 ± 70.8	0.346	0.399
Random	99.7 ± 10.9	0.222	0.215
Fixed idle (upper bound)	380.4	0.845	0.845

Table 3: Full benchmark results (50 seeds). Standard UCB at $c = 0.5$ reaches 94% of the always-idle ceiling. Contextual agents universally underperform their standard counterparts.

Key findings:

1. **Std UCB at $c = 0.5$ is dominant:** 358.1 total, 94% of the always-idle ceiling, with the lowest variance (14.0 vs 70–106 for others). Hyperparameter tuning is critical: the default $c = 2.0$ yielded only 240.4.
2. **Contextual agents are worse:** Every contextual variant underperforms its standard counterpart, the opposite of the MVP result. This suggests `step_bin` as a context feature misleads agents in the bootstrap dynamics – transitions are non-stationary (vary by timestep) and the current `step_bin` does not reliably predict intervention efficacy.
3. **The $c = 0.5$ UCB achieves consistency:** with ± 14.0 std dev vs ± 33.9 at $c = 2.0$, tuning simultaneously improves both mean and variance.

5 Masked Reward Benchmark

Same 9 agents on `configs/sprint1_bootstrap_extensions_masked.yaml` where step 4 (end of day) yields zero reward. Results are 50 seeds.

6 Comparison to Random-Transition Baseline

7 Key Findings

1. **Bootstrap transitions have exploitable structure.** Unlike the Dirichlet-random baselines where all agents performed similarly (~ 185 total), the LLM-estimated tables yield a 259% gap between best learner and random. The LLM produces non-trivial transition dynamics.

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	265.9 ± 37.3	0.591	0.681
Standard D-EG	248.3 ± 76.2	0.552	0.616
Standard EG ($\epsilon = 0.05$)	195.8 ± 98.4	0.435	0.518
Contextual UCB	178.4 ± 35.2	0.397	0.501
Standard TS	174.6 ± 70.6	0.388	0.456
Contextual D-EG	169.9 ± 70.0	0.378	0.435
Contextual EG	163.7 ± 73.9	0.364	0.443
Contextual TS	119.0 ± 38.5	0.264	0.289
Random	83.6 ± 8.4	0.186	0.179

Table 4: Masked benchmark results (50 seeds, step 4 zero-reward). Relative ordering is preserved. D-EG is most mask-resilient (86% retention).

Metric	Random Transitions	Bootstrap Transitions
Best agent	Std D-EG 189.1 (0.420)	Std UCB 358.1 (0.796)
Random baseline	183.5 (0.408)	99.7 (0.222)
Best vs random gap	3%	259%
Contextual advantage	None ($\approx 0\%$)	Negative (all worse)
Variance (best agent)	± 82	± 14

Table 5: Bootstrap transitions vs random transitions. The LLM-estimated tables produce structured dynamics with large agent differentials, while random transitions wash out all policy differences. Std UCB’s low variance under bootstrap suggests reliable convergence.

2. **Always-idle is near-optimal.** The LLM estimates that most users naturally maintain moderate-to-active step counts. Interventions incur a penalty without consistently improving outcomes, making inaction the best policy. Std UCB learns this implicitly, reaching 94% of the idle ceiling.
3. **Context features hurt.** Every contextual variant underperforms its standard counterpart, the opposite of the MVP result. This is likely because `step_bin` the next state depends weakly on current `step_bin` given the timestep-specific tables – context adds noise, not signal.
4. **UCB hyperparameter sensitivity is extreme.** Default $c = 2.0$: 240. Tuned $c = 0.5$: 358. The MVP-optimised values transferred well, but a dedicated grid search may find even better settings.
5. **The greedy oracle (one-step lookahead) is not a good bound** at 281, far below Std UCB’s 358. Myopic expected-reward maximisation fails because actions affect future states through burden accumulation and sleep at day boundaries.
6. **The LLM’s transition estimates are a plausible starting point** but should be validated against real data. The strong idle preference may reflect LLM bias toward inaction rather than true user dynamics.